

Natural Computing

LMU Munich
summer term 2025

Thomas Gabor





Werkstudent Softwareentwickler (m/w/d)

Quantum Computing, Web & KI

Deine Aufgaben:

- Software- & Schnittstellenentwicklung (REST-APIs, QC-Hardware)
- Frontend-Visualisierung mit React.js
- DevOps & Deployment (Docker, Kubernetes)
- Analyse & Entwurf einer Domänensprache (DSL)
- Hybride Algorithmen (Python, klassisch + quantenbasiert)

Dein Profil:

- Laufendes Studium IT/Physik/Mathe/Ing.
- Interesse Quantencomputing & Kommunikationstech.
- Erste Erfahrung Python/JavaScript, React.js+, REST, Docker/K8s von Vorteil

Wir bieten:

- ✓ 100% Remote
- ✓ Teilzeit
- ✓ Flexible Arbeitszeiten

Ansprechpartner:
Tobias Rohe



AIGNOSCO GmbH, München

Mind the calendar!

24	—	2025-06-12 Lecture 08
25	2025-06-17 Lecture 09	—
26	2025-06-24 Exercise 04	2025-06-26 Lecture 10

Exam Date

Tuesday, July 29, 2025

start 9:00, duration 90 minutes

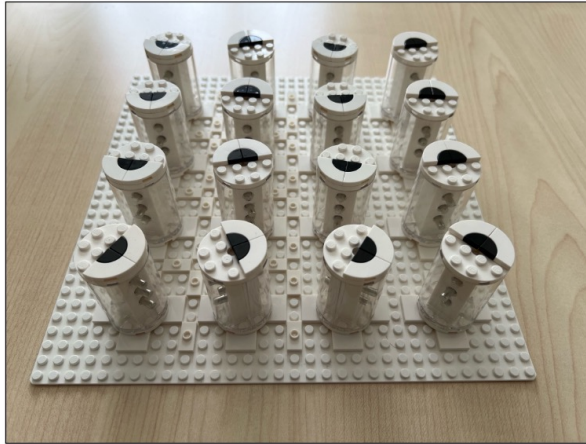
Großhaderner Str. 9, room N 00.001

details and registration soon in moodle

Previously on...



Experiment Time!



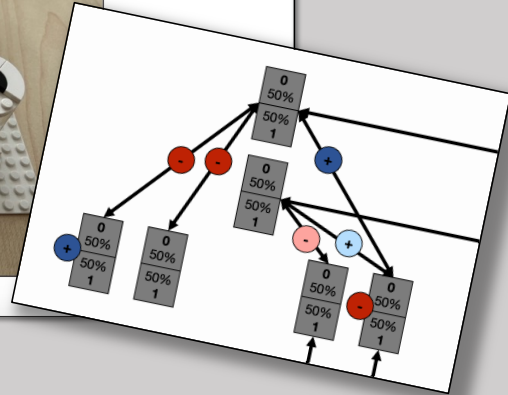
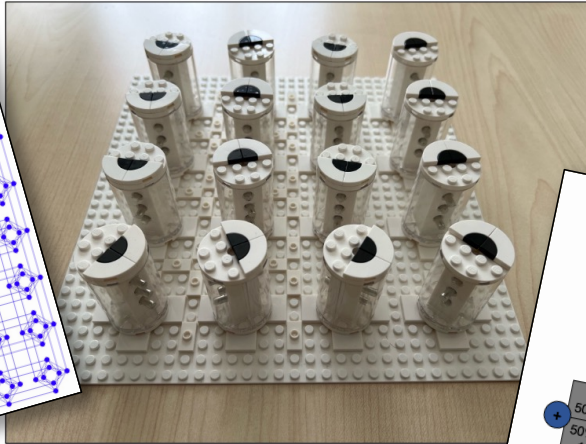
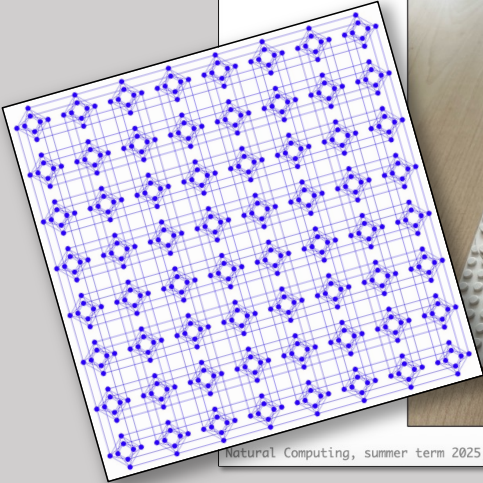
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Previously on...



Experiment Time!



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Definition 7 (Ising model). Let $G = (V, E)$ be a graph with vertices $V = \{v_0, \dots, v_{n-1}\}$ for $n \in \mathbb{N}$ and undirected edges $E \subseteq V \times V$. A spin configuration $x \in \mathcal{X} = (V \rightarrow \{-1, +1\})$ is a mapping of vertices to values $\{-1, +1\}$. W.l.o.g. we assume that x has the form $x(v_i) = \sigma_i$ for any $i \in \{0, \dots, n-1\}$ and a spin vector $\vec{\sigma} \in \{-1, +1\}^n$. The energy of an Ising model is then given via

$$H(\vec{\sigma}) = - \sum_{\substack{i \in \{0, \dots, n-1\} \\ j \in \{0, \dots, n-1\}}} J_{ij} \sigma_i \sigma_j - \sum_{i \in \{0, \dots, n-1\}} h_i \sigma_i$$

where $J \in \mathbb{R}^{n \times n}$ is called the interaction matrix and $\vec{h} \in \mathbb{R}^n$ is called the vector of field strengths. We also assume that for any $i, j \in \{0, \dots, n-1\}$ it holds that $i \geq j \implies J_{ij} = 0$, i.e., J is a strictly upper triangle matrix, and that $(v_i, v_j) \notin E \implies J_{ij} = 0$, i.e., J_{ij} can only have non-zero values where an edge between i and j exists in the substrate graph G .

For an Ising model given by (G, J, h) , there are processes (both artificial and natural) that produce a spin configuration $\vec{\sigma}$ so that $H(\vec{\sigma})$ is minimized.

Definition 8 (QUBO). Let $Q \in \mathbb{R}^{n \times n}$ be an arbitrary matrix called QUBO matrix. The process of finding a vector $\vec{x} \in \mathbb{B}^n$ so to

$$\text{minimize } H(\vec{x}) = x^\top Q x = \sum_{i=0}^{n-1} \sum_{j=0}^{n-1} Q_{ij} x_i x_j$$

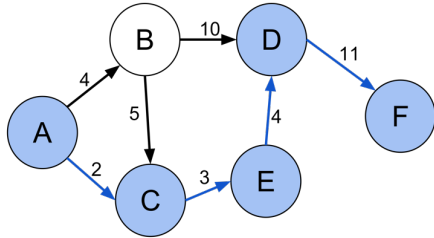
is called *quadratic unconstrained binary optimization (QUBO)*. W.l.o.g. we can assume that Q is a (non-strictly) upper triangular matrix.

Ising–QUBO Equivalence

Metaheuristic Search and Optimization

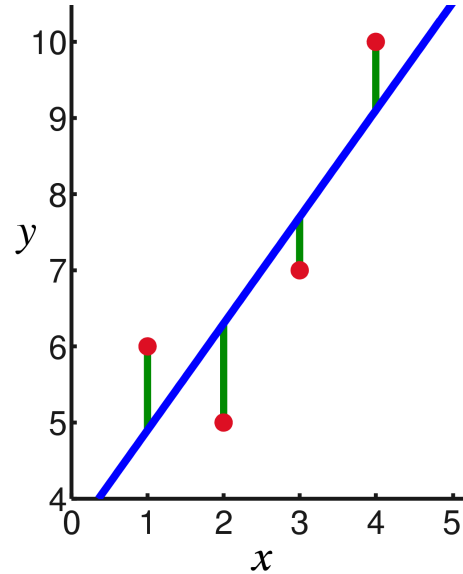
Definition 9 (optimization). Let $\mathcal{X}$ be an arbitrary state space. Let $\mathcal{T}$ be an arbitrary set called target space and let $\leq$ be a total order on $\mathcal{T}$. A total function $\tau : \mathcal{X} \rightarrow \mathcal{T}$ is called target function. Optimization (minimization/maximization) is the procedure of searching for an $x \in \mathcal{X}$ so that $\tau(x)$ is optimal (minimal/maximal). Unless stated otherwise, we assume minimization. An optimization run of length $g + 1$ is a sequence of states $\langle x_t \rangle_{0 \leq t \leq g}$ with $x_t \in \mathcal{X}$ for all t .

Shortest Path



source:
[en.wikipedia.org/wiki/Shortest_path_problem#/media/
File:Shortest_path_with_direct_weights.svg](https://en.wikipedia.org/wiki/Shortest_path_problem#/media/File:Shortest_path_with_direct_weights.svg)

Linear Regression



source:
[en.wikipedia.org/wiki/Linear_regression#/media/
File:Linear_least_squares_example2.png](https://en.wikipedia.org/wiki/Linear_regression#/media/File:Linear_least_squares_example2.png)

Eierkartons!



source: Wikipedia

Wir brauchen Eierkartons...